

Derivación Semántica Formal – Notación Nielson

Rama 1 de 2 (SOS)

$$\begin{aligned}\mathcal{T}_1 &= \langle x := 1, \sigma_0 \rangle \Rightarrow \sigma_0[x \mapsto \mathcal{A}[[1]]\sigma_0] \text{ [ass}_{\text{sos}}] \\ \mathcal{T}_2 &= \frac{\mathcal{T}_1}{\langle S_1, \sigma_0 \rangle \Rightarrow \langle \langle x := 0 ; x := x + 2 \rangle, \sigma_1 \rangle} \text{ [par}_{\text{sos}}^2] \\ \mathcal{T}_3 &= \langle x := 0, \sigma_1 \rangle \Rightarrow \sigma_1[x \mapsto \mathcal{A}[[0]]\sigma_1] \text{ [ass}_{\text{sos}}] \\ \mathcal{T}_4 &= \frac{\mathcal{T}_3}{\langle x := 0 ; x := x + 2, \sigma_1 \rangle \Rightarrow \langle x := x + 2, \sigma_2 \rangle} \text{ [comp}_{\text{sos}}^2] \\ \mathcal{T}_5 &= \langle x := x + 2, \sigma_2 \rangle \Rightarrow \sigma_2[x \mapsto \mathcal{A}[[x + 2]]\sigma_2] \text{ [ass}_{\text{sos}}] \\ \mathcal{T}_6 &= \frac{\mathcal{T}_4, \mathcal{T}_5}{\langle \langle x := 0 ; x := x + 2 \rangle, \sigma_1 \rangle \Rightarrow \sigma_3} \text{ [protect}_{\text{sos}}]\end{aligned}$$

Sentencias Abreviadas

$$S_1 = x := 1 \parallel \langle x := 0 ; x := x + 2 \rangle \quad (\mathbf{S}_{\text{ini}})$$

Leyenda de Estados

- $\sigma_0 \equiv \emptyset$ (σ_{ini})
- $\sigma_1 \equiv \sigma_0[x \mapsto \mathcal{A}[[1]]\sigma_0] = \{x \mapsto 1\}$
- $\sigma_2 \equiv \sigma_1[x \mapsto \mathcal{A}[[0]]\sigma_1] = \{x \mapsto 0\}$
- $\sigma_3 \equiv \sigma_2[x \mapsto \mathcal{A}[[x + 2]]\sigma_2] = \{x \mapsto 2\}$ (σ_{fin})

Rama 2 de 2 (SOS)

$$\begin{aligned}\mathcal{T}_1 &= \langle x := 0, \sigma_0 \rangle \Rightarrow \sigma_0[x \mapsto \mathcal{A}[[0]]\sigma_0] \text{ [ass}_{\text{sos}}] \\ \mathcal{T}_2 &= \frac{\mathcal{T}_1}{\langle x := 0 ; x := x + 2, \sigma_0 \rangle \Rightarrow \langle x := x + 2, \sigma_1 \rangle} \text{ [comp}_{\text{sos}}^2] \\ \mathcal{T}_3 &= \langle x := x + 2, \sigma_1 \rangle \Rightarrow \sigma_1[x \mapsto \mathcal{A}[[x + 2]]\sigma_1] \text{ [ass}_{\text{sos}}] \\ \mathcal{T}_4 &= \frac{\mathcal{T}_2, \mathcal{T}_3}{\langle \langle x := 0 ; x := x + 2 \rangle, \sigma_0 \rangle \Rightarrow \sigma_2} \text{ [protect}_{\text{sos}}] \\ \mathcal{T}_5 &= \frac{\mathcal{T}_4}{\langle S_1, \sigma_0 \rangle \Rightarrow \langle x := 1, \sigma_2 \rangle} \text{ [par}_{\text{sos}}^4] \\ \mathcal{T}_6 &= \langle x := 1, \sigma_2 \rangle \Rightarrow \sigma_2[x \mapsto \mathcal{A}[[1]]\sigma_2] \text{ [ass}_{\text{sos}}]\end{aligned}$$

Sentencias Abreviadas

$$S_1 = x := 1 \parallel \langle x := 0 ; x := x + 2 \rangle \quad (\mathbf{S}_{\text{ini}})$$

Leyenda de Estados

- $\sigma_0 \equiv \varnothing$ (σ_{ini})
- $\sigma_1 \equiv \sigma_0[x \mapsto \mathcal{A}[\![0]\!]\sigma_0] = \{x \mapsto 0\}$

- $\sigma_2 \equiv \sigma_1[x \mapsto \mathcal{A}[\![x + 2]\!]\sigma_1] = \{x \mapsto 2\}$
- $\sigma_3 \equiv \sigma_2[x \mapsto \mathcal{A}[\![1]\!]\sigma_2] = \{x \mapsto 1\}$ (σ_{fin})